Reg. No. | | | | | | | | | |

Question Paper Code : 17976

B.E. / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2021

Third Semester

B.E. – Computer Science and Engineering

19CS301 – DATA STRUCTURES AND ALGORITHMS

(Regulations: Mepco – R2019)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

PART A – (10 × 2 = 20 Marks)

BTL, CO

R, CO1 1. Write down the steps involved in analyzing a recursive algorithm.

A, CO1 2. What is the time complexity of the below function? Justify your answer.

```
void fun(int n, int arr[])
{
  int i = 0;
  for(; i < n; ++i)
  {
     j = 0;
     while(j < n && arr[i] < arr[j])
     {
        j++;
     }
  }
}
```

A) O(n) B) O(n^2) C) O(nlogn) D) O(n(logn)^2)

U, CO2 3. What are the limitations of singly linked list?

A, CO2 4. The data structure required for Depth First Traversal on a graph is_____. Justify your answer.

A) Stack B) Queue C) Tree D) Array

U, CO3 5. What is a splay tree?

A, CO3 6. The following code will be used to find _____in a Binary Search tree. Justify your answer.

```
public void displayValue(Tree root)
{
        while(root.left() != null)
        {
                root = root.left();
        }
        System.out.println(root.data());
 }
```

A) Max Element B) Min Element C) Empty node D) Root node

U, CO4 7. What is an Equivalence relation?

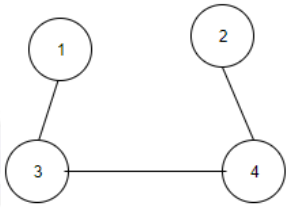
U, CO4 8. Which of the following collision resolution strategy is used for avoiding secondary clustering problem? Justify your answer.

A) Linear probing B) Quadratic probing

C) Double hashing D) Separate chaining

R, CO5 9. Define Euler circuit.

A, CO5 10. What would be the number of zeros in the adjacency matrix of the given graph? Justify your answer.



A) 10 B) 6 C) 0 D) 16

PART B – (5 × 16 = 80 Marks)

U, CO1 11. a) i. Explain how recursive algorithms are different from non-recursive algorithms while performing binary search. (8 Marks)

A, CO1 11. a) ii. Solve the following recurrences by Backward substitution method: $x(n) = x(n-1) + 5$ for $n > 1$, $x(1) = 0$ (8 Marks)

OR

U, CO1 11. b) i. Explain the importance of asymptotic notations in the analysis of algorithm. Explain with big O and θ notations. (8 Marks)

A, CO1 11. b) ii. Write a non-recursive algorithm for Matrix Addition and analyze the time complexity of the algorithm. (8 Marks)

U, CO2 12. a) i. Design an ADT for student workshop registration where the students are added to the workshop on first come first serve basis. Write an algorithm to implement the same using linked list. (8 Marks)

U, CO2 12. a) ii. Write an algorithm to evaluate a postfix expression using stack. Evaluate the following postfix expression:

ABCD^*E-*F*G/

Let A=5, B=3, C=2, D=3, E=4, F=5, G=10 (8 Marks)

OR

U, CO2 12. b) i. List down the limitations of implementing the linear queue using arrays. Explain how circular queue can be used to overcome the problem and write an algorithm for enqueue and dequeue in a circular queue using arrays. (8 Marks)

U, CO2 12. b) ii. Write an algorithm to perform the following operation on list using doubly linked list:

A) Insert after an element

B) Delete front

C) Display (8 Marks)

U, CO3 13. a) i. Explain about the various rotations involved in AVL tree insertion with an example. (8 Marks)

A, CO3 13. a) ii. Define MinHeap. State the properties associated with MinHeap. Perform the following operations on a MinHeap:

A) Insert 25, 40, 32, 12, 60, 30, 50, 35, 20, 55

B) Perform deleteMin two times

Show the tree at each stage with proper node structure. (8 Marks)

OR

U, CO3 13. b) i. Explain in detail about the various rotation performed in a splay tree with an example. (8 Marks)

A, CO3 13. b) ii. What is a Binary Search Tree? Perform the following operations on a Binary Search Tree:

A) Insert 67,34,89,45,12,93,75,25,38,56 into an empty tree

B) Delete 12, 89, 67

Show the tree at each stage with proper node structure. (8 Marks)

U, CO4 14. a) i. Write an algorithm for performing union and find operation in a simple disjoint set. (8 Marks)

A, CO4 14. a) ii. Define Closed Hashing. Using linear probing insert the following elements in the hash table: 25,39,45,67,89,98,21,33,19,76

After insertion **Delete 67** then I**nsert 12**

Note: Table size has to be chosen by the designer (8 Marks)

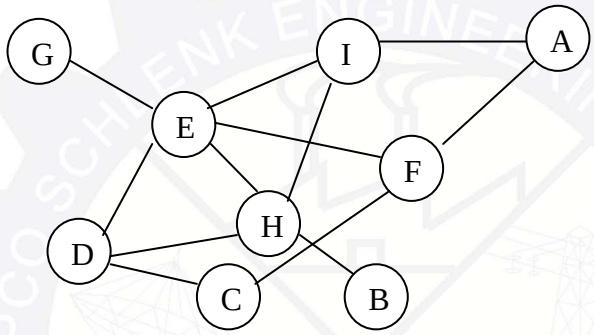
OR

U, CO4 14. b) i. Define primary clustering. Explain how quadratic probing can be used to avoid primary clustering with an algorithm. (8 Marks)

A, CO4 14. b) ii. Show step by step result of the following sequence of instructions when unions are performed by size:
union(1,2), union(3,4), union(3,5), union(1,7), union(3,6), union(8,9), union(1,8), union(3,10), union(3,11), union(3,12) (8 Marks)

U, CO5 15. a) i. Define connected components. Devise an algorithm to find the connected components of the graph. (8 Marks)

A, CO5 15. a) ii. Perform breadth first traversal in the given graph starting from vertex A:

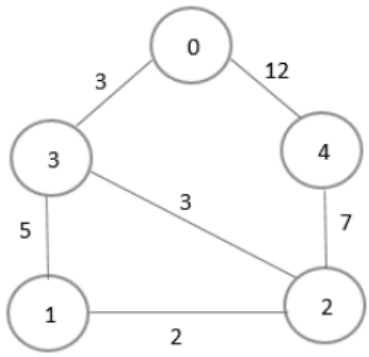


(8 Marks)

OR

U, CO5 15. b) i. Write an algorithm for topological sorting in a graph. (8 Marks)

A, CO5 15. b) ii. Define Minimum Spanning Tree. Find the minimum spanning tree for the following graph using Kruskal's algorithm.



(8 Marks)